

MID-TERM EXAMINATION

GENERAL SCIENCE, MATHEMATICS & COMPUTING

Max Marks: 100

SECTION A: MATHEMATICS

1. A trader buys a school bag for ₹800 and sells it for a profit of 20%. Find the selling price. [5 Marks]
8. Find the area of a triangle with a base of 14 cm and a height of 10 cm. [4 Marks]

SECTION B: BIOLOGY

2. Draw and label a diagram of an **animal cell**, showing the cell membrane, nucleus, cytoplasm, and mitochondria. [8 Marks]
9. Draw a labeled diagram showing the **human digestive system**, including the mouth, oesophagus, stomach, small intestine, and large intestine. [8 Marks]

SECTION C: PHYSICS

3. (a) Derive the second equation of motion, $s = ut + \frac{1}{2}at^2$, using the first equation of motion. [5 Marks]
- (b) A motorcycle starts from rest and reaches a velocity of 20 m/s in 5 seconds. Calculate its acceleration. [5 Marks]

SECTION D: GENERAL KNOWLEDGE

4. Which is the smallest planet in our solar system? [2 Marks]
10. Who was the first President of India? [2 Marks]

SECTION E: CHEMISTRY

5. (a) Draw the structure of **ethanol (C₂H₅OH)** and name its functional group. [4 Marks]
- (b) Name the group and period of **Magnesium (Mg)** in the periodic table and state one physical property of magnesium. [4 Marks]

SECTION F: COMPUTER SCIENCE

6. Write a Python program to check whether a given number is **even or odd**. [6 Marks]

SECTION G: THEORY

7. (a) Explain **Newton's First Law of Motion** using a short story or a real-life example. [5 Marks]
- (b) Explain the **carbon cycle** using a short story or real-life analogy. [5 Marks]

[END OF QUESTION PAPER]